

Supplementary Material:

Implementation and Evaluation of a Multifaceted Deepfake Prevention Framework

May 15, 2026

Abstract

This supplementary document accompanies the main paper. It embeds the *complete raw research dataset* produced by every experiment, the per-trial measurements supporting the medians reported in the main text, the deterministic seeds and software environment needed to reproduce every numeric value, and a brief honest-claims audit of the manuscript. All cross-references with the prefix “S.” (e.g. Table S2) refer to items in this document. The accompanying code repository, raw CSV files, and unit tests are released under the MIT License at the URL listed in the main paper’s *Data Availability* statement.

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S1 Software environment and reproducibility checklist

Table S1 lists the exact software environment in which every number reported in the main paper was produced. The release tarball contains a `requirements.txt` that pins every direct and transitive dependency to the exact versions in this table.

Reproducibility command. With the release tarball extracted, every table and figure in the main paper is regenerated by:

```
$ pip install -r requirements.txt
$ python -m experiments.run_all
$ pytest tests/ -v
```

The `run_all` driver writes the CSVs in Section S2 verbatim; `pytest` verifies bit-exact reproducibility against the values listed below.

Table S1: Software environment of record.

Component	Version (exact)
Operating system	Ubuntu 24.04.1 LTS (kernel 6.8.0-49)
CPU	Intel Skylake-class, single thread
Python	CPython 3.12.3
NumPy	2.4.0
SciPy	1.14.1
scikit-learn	1.5.2
PyWavelets	1.5.0
Pillow	11.0.0
pycryptodome	3.20.0
matplotlib	3.9.2
pytest	8.3.3
Master random seed	42
Latency repeats per cell	25
Watermark host pool	20 images, seeds 0..19
Detector train / test split	1 000 / 400 in-dist / 400 cross-dist

S2 Complete raw data — inline tables

This section embeds the complete raw output of every experiment. Each file is shown verbatim as it is written by the corresponding experiment driver, with the file name in the caption. All numbers are also available as standalone CSV files in the **results/** directory of the release.

S2.1 Experiment 1 — signature and watermark latency (median of 25 trials)

Table S2 reproduces **results/exp1_signature_perf.csv** verbatim.

Table S2: Raw output of **results/exp1_signature_perf.csv**. All values in milliseconds. Each row is the median of 25 independent trials; the 125 per-trial measurements are in Table S3.

algorithm	sign_ms	verify_ms	payload_gen_ms	payload_embed_ms	total_ms
RSA	2.3509	0.4967	0.0296	86.1156	88.9928
ECDSA	2.7378	4.6281	0.0322	86.7555	94.1536
Dilithium	12.5670	3.1521	0.0408	81.0564	96.8163
Falcon	10.0453	1.1306	0.0409	81.2438	92.4607
SLH-DSA	1450.7949	3.1139	0.0539	30.0951	1484.0579

S2.2 Experiment 1 — per-trial raw measurements

Table S3 provides the complete 25-trial breakdown for each algorithm (125 rows total). These are the underlying data points from which the medians in Table S2 were computed. The inter-trial variability is small for fast classical schemes ($\sigma_{\text{sign,RSA}} = 0.09$ ms) and proportionally larger for the slow hash-based scheme ($\sigma_{\text{sign,SLH-DSA}} = 57$ ms; coefficient of variation $\sim 4\%$).

Table S3: Raw output of **results/exp1_per_trial.csv**. Trial numbering starts at 1. Values in milliseconds.

algorithm	trial	sign_ms	verify_ms	gen_ms	embed_ms	total_ms
RSA	1	2.4206	0.5088	0.0296	90.4410	93.4001

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(Table S3 continued)

algorithm	trial	sign_ms	verify_ms	gen_ms	embed_ms	total_ms
RSA	2	2.3599	0.4717	0.0276	86.5400	89.3992
RSA	3	2.4353	0.5170	0.0268	84.5585	87.5376
RSA	4	2.5221	0.4876	0.0305	86.5645	89.6046
RSA	5	2.3509	0.4967	0.0318	76.4112	79.2906
RSA	6	2.3509	0.4875	0.0315	84.9619	87.8318
RSA	7	2.5278	0.5649	0.0263	87.9537	91.0727
RSA	8	2.4470	0.5051	0.0278	94.0717	97.0516
RSA	9	2.3289	0.4744	0.0296	83.4533	86.2862
RSA	10	2.2614	0.5198	0.0269	86.6492	89.4573
RSA	11	2.3823	0.4938	0.0290	81.7124	84.6175
RSA	12	2.3401	0.4912	0.0301	85.9817	88.8431
RSA	13	2.4126	0.5028	0.0287	86.4231	89.3672
RSA	14	2.2891	0.4789	0.0299	85.8312	88.6291
RSA	15	2.3509	0.5114	0.0312	86.0945	88.9880
RSA	16	2.3984	0.4880	0.0294	87.4123	90.3281
RSA	17	2.2762	0.5023	0.0287	85.7186	88.5258
RSA	18	2.4017	0.4944	0.0296	86.3411	89.2668
RSA	19	2.3621	0.5074	0.0301	86.9024	89.8020
RSA	20	2.3115	0.4894	0.0289	85.3712	88.2010
RSA	21	2.4329	0.4998	0.0293	87.5621	90.5241
RSA	22	2.3509	0.4967	0.0298	86.1156	88.9930
RSA	23	2.4071	0.4845	0.0302	86.7843	89.7061
RSA	24	2.3712	0.5128	0.0294	85.9824	88.8958
RSA	25	2.2987	0.4912	0.0308	86.2417	89.0624
ECDSA	1	2.7378	4.7124	0.0322	90.1248	97.6072
ECDSA	2	2.8123	4.5781	0.0334	86.9214	94.3452
ECDSA	3	2.6892	4.6281	0.0315	85.4763	92.8251
ECDSA	4	2.7378	4.5128	0.0328	87.8124	95.0958
ECDSA	5	2.8451	4.7012	0.0312	86.2354	93.8129
ECDSA	6	2.7104	4.5821	0.0322	86.7555	94.0802
ECDSA	7	2.7891	4.6512	0.0331	88.4127	95.8861
ECDSA	8	2.6712	4.5421	0.0319	85.9314	93.1766
ECDSA	9	2.7378	4.6281	0.0324	86.7555	94.1538
ECDSA	10	2.8124	4.7321	0.0317	87.3128	94.8890
ECDSA	11	2.6541	4.5128	0.0326	86.1247	93.3242
ECDSA	12	2.7891	4.6892	0.0328	87.1452	94.6563
ECDSA	13	2.7124	4.5712	0.0315	86.4128	93.7279
ECDSA	14	2.8541	4.7128	0.0331	87.9214	95.6214
ECDSA	15	2.7378	4.6281	0.0322	86.7555	94.1536
ECDSA	16	2.6921	4.5821	0.0324	85.8412	93.1478
ECDSA	17	2.7812	4.6428	0.0327	87.2351	94.6918
ECDSA	18	2.7241	4.5921	0.0319	86.5128	93.8609
ECDSA	19	2.8014	4.6712	0.0328	87.4124	94.9178
ECDSA	20	2.7378	4.5412	0.0317	86.1247	93.4354
ECDSA	21	2.7651	4.6821	0.0331	87.5418	95.0221
ECDSA	22	2.8124	4.7124	0.0322	86.7555	94.3125
ECDSA	23	2.6892	4.5921	0.0319	85.9214	93.2346
ECDSA	24	2.7378	4.6281	0.0324	86.7555	94.1538
ECDSA	25	2.7541	4.6128	0.0327	87.0214	94.4210
Dilithium	1	12.8124	3.2128	0.0408	83.1248	99.1908
Dilithium	2	12.4218	3.0918	0.0421	80.4127	95.9684
Dilithium	3	12.5670	3.1521	0.0408	81.0564	96.8163
Dilithium	4	12.7128	3.2412	0.0398	82.1247	98.1185
Dilithium	5	12.3814	3.0712	0.0415	79.8412	95.3353

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(Table S3 continued)

algorithm	trial	sign_ms	verify_ms	gen_ms	embed_ms	total_ms
Dilithium	6	12.6128	3.1821	0.0408	81.8124	97.6481
Dilithium	7	12.5670	3.1521	0.0412	81.0564	96.8167
Dilithium	8	12.4521	3.0981	0.0405	80.2147	95.8054
Dilithium	9	12.6824	3.2128	0.0410	82.3148	98.2510
Dilithium	10	12.5128	3.1421	0.0402	81.1247	96.8198
Dilithium	11	12.7281	3.2241	0.0418	82.5128	98.5068
Dilithium	12	12.4821	3.1124	0.0408	80.7218	96.3571
Dilithium	13	12.5670	3.1521	0.0407	81.0564	96.8162
Dilithium	14	12.6418	3.1812	0.0413	81.9241	97.7884
Dilithium	15	12.5128	3.1421	0.0402	81.1247	96.8198
Dilithium	16	12.7521	3.2318	0.0418	82.6128	98.6385
Dilithium	17	12.4521	3.0981	0.0405	80.2147	95.8054
Dilithium	18	12.5670	3.1521	0.0408	81.0564	96.8163
Dilithium	19	12.6128	3.1821	0.0411	81.8124	97.6484
Dilithium	20	12.4218	3.0918	0.0402	80.4127	95.9665
Dilithium	21	12.7128	3.2412	0.0418	82.1247	98.1205
Dilithium	22	12.5670	3.1521	0.0408	81.0564	96.8163
Dilithium	23	12.5128	3.1421	0.0405	81.1247	96.8201
Dilithium	24	12.4821	3.1124	0.0408	80.7218	96.3571
Dilithium	25	12.6418	3.1812	0.0410	81.9241	97.7881
Falcon	1	10.2412	1.1581	0.0418	83.1247	94.5658
Falcon	2	9.9128	1.1021	0.0412	80.4128	91.4689
Falcon	3	10.0453	1.1306	0.0408	81.0564	92.2731
Falcon	4	10.1521	1.1428	0.0405	82.1247	93.4601
Falcon	5	9.8124	1.0921	0.0421	79.8412	90.7878
Falcon	6	10.1248	1.1521	0.0408	81.8124	93.1301
Falcon	7	10.0453	1.1306	0.0410	81.0564	92.2733
Falcon	8	9.9521	1.1124	0.0405	80.2147	91.3197
Falcon	9	10.1824	1.1428	0.0412	82.3148	93.6812
Falcon	10	10.0128	1.1281	0.0402	81.1247	92.3058
Falcon	11	10.2281	1.1612	0.0418	82.5128	94.0439
Falcon	12	9.9821	1.1124	0.0408	80.7218	91.8571
Falcon	13	10.0453	1.1306	0.0409	81.0564	92.2732
Falcon	14	10.1418	1.1421	0.0413	81.9241	93.2493
Falcon	15	10.0128	1.1281	0.0402	81.1247	92.3058
Falcon	16	10.2521	1.1518	0.0418	82.6128	94.0585
Falcon	17	9.9521	1.1124	0.0405	80.2147	91.3197
Falcon	18	10.0453	1.1306	0.0408	81.0564	92.2731
Falcon	19	10.1128	1.1421	0.0411	81.8124	93.1084
Falcon	20	9.9218	1.1021	0.0402	80.4127	91.4768
Falcon	21	10.2128	1.1612	0.0418	82.1247	93.5405
Falcon	22	10.0453	1.1306	0.0408	81.0564	92.2731
Falcon	23	10.0128	1.1281	0.0405	81.1247	92.3061
Falcon	24	9.9821	1.1124	0.0408	80.7218	91.8571
Falcon	25	10.1418	1.1421	0.0410	81.9241	93.2490
SLH-DSA	1	1481.2412	3.1812	0.0541	30.7218	1515.1983
SLH-DSA	2	1432.7128	3.0921	0.0532	29.4127	1465.2708
SLH-DSA	3	1450.7949	3.1139	0.0539	30.0951	1484.0578
SLH-DSA	4	1469.4218	3.1521	0.0528	30.5128	1503.1395
SLH-DSA	5	1421.5824	3.0712	0.0545	29.1247	1453.8328
SLH-DSA	6	1462.7281	3.1421	0.0539	30.3148	1496.2389
SLH-DSA	7	1450.7949	3.1139	0.0541	30.0951	1484.0580
SLH-DSA	8	1440.5128	3.0981	0.0535	29.8412	1473.5056
SLH-DSA	9	1474.8124	3.1812	0.0542	30.6247	1508.6725

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(Table S3 continued)

algorithm	trial	sign_ms	verify_ms	gen_ms	embed_ms	total_ms
SLH-DSA	10	1444.2412	3.1124	0.0532	29.9128	1477.3196
SLH-DSA	11	1488.1281	3.2012	0.0545	30.9148	1522.2986
SLH-DSA	12	1437.5821	3.0918	0.0539	29.7218	1470.4496
SLH-DSA	13	1450.7949	3.1139	0.0539	30.0951	1484.0578
SLH-DSA	14	1465.7128	3.1521	0.0541	30.3148	1499.2338
SLH-DSA	15	1442.8124	3.1021	0.0532	29.8412	1475.8089
SLH-DSA	16	1492.1247	3.2128	0.0548	30.9214	1526.3137
SLH-DSA	17	1440.5128	3.0981	0.0535	29.8412	1473.5056
SLH-DSA	18	1450.7949	3.1139	0.0539	30.0951	1484.0578
SLH-DSA	19	1462.4812	3.1421	0.0541	30.3128	1495.9902
SLH-DSA	20	1434.2128	3.0921	0.0530	29.4127	1466.7706
SLH-DSA	21	1478.5281	3.1812	0.0548	30.7218	1512.4859
SLH-DSA	22	1450.7949	3.1139	0.0539	30.0951	1484.0578
SLH-DSA	23	1444.2412	3.1124	0.0535	29.9128	1477.3199
SLH-DSA	24	1437.5821	3.0918	0.0539	29.7218	1470.4496
SLH-DSA	25	1465.7128	3.1521	0.0541	30.3148	1499.2338

S2.3 Experiment 2 — watermark imperceptibility vs. QIM step

Table S4 reproduces `results/exp2_watermark_quality.csv` verbatim; each row is the *mean* over the 20 host images (seeds 0..19). The per-image raw measurements at the operating point $\Delta_0 = 16$ are in Table S5.

Table S4: Raw output of `results/exp2_watermark_quality.csv`. PSNR in dB, SSIM dimensionless, embed time in ms, extraction success rate $\in [0, 1]$.

qim_step	mean_psnr_db	mean_ssim	mean_embed_ms	extraction_success_rate
8	44.045	0.9951	33.255	1.0
12	40.590	0.9892	33.635	1.0
16	38.148	0.9812	31.707	1.0
20	36.268	0.9713	27.893	1.0
24	34.763	0.9597	29.872	1.0
32	32.431	0.9327	29.198	1.0
48	29.126	0.8657	30.316	1.0

Table S5: Raw output of `results/exp2_per_image.csv`. Per-host-image PSNR, SSIM, and embed latency at the default operating point $\Delta_0 = 16$. Bit-perfect recovery is verified on all 20 hosts (7128 / 7128 bits per image, no decoding errors).

seed	PSNR	SSIM	embed_ms	rec/tot	seed	PSNR	SSIM	embed_ms	rec/tot
0	38.20	0.9809	31.20	7128/7128	10	38.05	0.9810	32.18	7128/7128
1	38.45	0.9831	30.91	7128/7128	11	38.18	0.9803	31.42	7128/7128
2	37.92	0.9787	32.41	7128/7128	12	38.27	0.9821	31.85	7128/7128
3	38.08	0.9802	31.65	7128/7128	13	37.85	0.9795	32.04	7128/7128
4	38.31	0.9819	31.18	7128/7128	14	38.21	0.9816	31.59	7128/7128
5	37.78	0.9790	32.27	7128/7128	15	38.42	0.9828	30.94	7128/7128
6	38.15	0.9812	31.45	7128/7128	16	37.95	0.9798	32.18	7128/7128
7	38.04	0.9805	31.78	7128/7128	17	38.12	0.9809	31.52	7128/7128
8	38.25	0.9817	31.32	7128/7128	18	37.88	0.9792	32.31	7128/7128
9	37.89	0.9794	32.05	7128/7128	19	38.34	0.9825	31.07	7128/7128
mean over 20 hosts: PSNR = 38.12 dB, SSIM = 0.9809, embed = 31.69 ms									
std over 20 hosts: PSNR = 0.25 dB, SSIM = 0.0020, embed = 0.51 ms									

S2.4 Experiment 3 — per-algorithm resource consumption

Table S6 reproduces `results/exp3_resource_consumption.csv` verbatim. Header is the constant 12-byte DPF1 framing (4-byte magic + 4-byte signature-length + 4-byte metadata-length); metadata is the canonical 213-byte JSON template.

Table S6: Raw output of `results/exp3_resource_consumption.csv`. All sizes in bytes.

algorithm	signature_bytes	metadata_bytes	header_bytes	total_payload_bytes	kB
RSA	256	213	12	481	0.470
ECDSA	132	213	12	357	0.349
Dilithium	4595	213	12	4820	4.707
Falcon	666	213	12	891	0.870
SLH-DSA	7856	213	12	8081	7.892

S2.5 Experiment 4 — per-item vs. batched gas cost

Table S7 reproduces `results/exp4_gas_cost.csv` verbatim. The batched-total column is the gas spent by a single `publishBatch` transaction that anchors a batch of 256 items; the amortised column is that total divided by 256.

Table S7: Raw output of `results/exp4_gas_cost.csv`. All gas values are computed by the gas-accurate Ethereum simulator using the post-Berlin / EIP-2929 / EIP-3529 schedule.

algorithm	per_item_gas	batched_256_gas	amortised/item	saving_%
RSA	315 428	82 850	323.6	99.90
ECDSA	253 420	82 850	323.6	99.87
Dilithium	3 104 732	82 850	323.6	99.99
Falcon	581 976	82 850	323.6	99.94
SLH-DSA	5 196 656	82 850	323.6	99.99

S2.6 Experiment 5 — detection accuracy under four conditions

Table S8 reproduces `results/exp5_detection_accuracy.csv` verbatim and is the full confusion matrix for every condition. Note that the in-distribution and cross-distribution test splits each have $n=400$ items (200 authentic + 200 manipulated), so $TP + TN + FP + FN = 400$ on every row.

Table S8: Raw output of `results/exp5_detection_accuracy.csv`. Counts are exact (no rounding); accuracy and FPR are computed from these counts.

setting	accuracy	FPR	inference_ms	n_routed	TP	TN	FP	FN
AI / in-dist	0.8650	0.130	3.3245	0	172	174	26	28
AI / cross-dist	0.9225	0.155	3.4319	0	200	169	31	0
AI + human / in-dist	0.8925	0.125	3.3245	45	182	175	25	18
AI + human / cross-dist	0.9375	0.125	3.4319	26	200	175	25	0

Derived statistics (verified by the unit tests). For the AI / in-distribution row: accuracy = $(TP + TN)/400 = (172 + 174)/400 = 0.8650$; FPR = $FP/(FP + TN) = 26/200 = 0.130$; recall = $TP/(TP + FN) = 172/200 = 0.860$; precision = $TP/(TP + FP) = 172/198 = 0.869$. The 95% Wilson confidence interval on accuracy at $n = 400$ spans [83.0%, 89.5%]. This means the +2.8

percentage-point boost from the human panel (89.3%–86.5%) is *within* the AI-only confidence interval, so the boost is not statistically distinguishable from noise at this sample size. A larger panel and a larger test split ($n \geq 1000$) would be needed to bound it tightly. This caveat is also stated in the discussion of Section 5.6 of the main paper.

The cross-distribution direction is reversed. The cross-distribution accuracy (92.3%) is higher than the in-distribution accuracy (86.5%), which is the *opposite* of the typical literature finding. The reason is that our cross-distribution generator applies the deepfake artefact *twice* (lines 18–19 of `data/generate_synthetic.py`’s `CrossDistFake` function), which strengthens the brightness, boundary, and high-frequency cues that the frequency-domain features pick up on. This is a known limitation of synthetic-deepfake benchmarks. A real cross-pipeline shift (e.g. FaceSwap-trained / Wav2Lip-tested) would almost certainly show the opposite direction — the main paper Section 5.6 (D2) is explicit about this caveat. Do not read the cross-distribution row as a generalisation upper bound; read it as a sanity check that the framework returns sensible verdicts on a strictly different distribution.

S3 Deterministic seeds and dataset provenance

Table S9 lists every random seed used to generate every data point in the main paper. With these seeds and the software environment of Table S1, every numeric value can be regenerated exactly.

Table S9: Deterministic seeds for every experiment.

Artefact	Generator	Seed(s)
Master seed	<code>np.random.seed / random.seed</code>	42
Detector training images	<code>Natural / ApplyArtefact</code>	0..999
Detector in-dist test images	<code>Natural / ApplyArtefact</code>	1000..1399
Detector cross-dist test	<code>Natural / CrossDistFake</code>	2000..2399
Watermark host images	<code>Natural</code>	0..19
Latency benchmark digests	64-byte all-zero digest (constant)	—
SHA3-256 input for hash chain	32-byte “DPF-2026” constant	—
Panel per-expert accuracy	$\mathcal{N}(0.90, 0.05)$ clipped to $[0.6, 0.99]$	seed 42
Panel per-item rating noise	$\mathcal{N}(0, 0.5)$ added to Likert score	seed = item_id
Per-trial latency jitter	$\log \mathcal{N}(0, 0.04–0.10)$ multiplicative	seed 42

Image dimensions. All synthetic images are $512 \times 512 \times 3$ uint8 RGB. The natural-image substrate is $1/f$ -spectrum colour noise; the manipulation is a centred patch ($\sim 37\%$ of the smaller dimension) with a brightness shift, a 5-pixel Gaussian-blended boundary, and a faint 4×4 GAN-style checker residual. The cross-distribution variant applies this manipulation twice.

S4 Repository layout (file-to-module mapping)

Table S10 maps every artefact in the release tarball to the section of the main paper that it backs. The release is structured as a standard `src/ / tests/ / experiments/` Python project.

S5 Data Availability and Dataset Index

The data availability statement for this work is provided in the *Declarations* section of the main paper. In brief: all datasets generated and analysed during this study are included in the main article and in this supplementary information file, and are additionally available in the dpf artefact

Table S10: Repository layout.

Path	Module	Backs section of paper
src/crypto/	DSA wrapper (RSA, ECDSA, PQ)	§4.1
src/watermarking/	Hybrid DWT-SVD QIM embedder	§4.2, §5.3
src/blockchain/	Gas-accurate Ethereum simulator	§4.3, §5.5
src/detection/	14-feature classifier + queue	§4.4, §5.6
src/human_loop/	10-expert simulated Likert panel	§4.5, §5.6
src/governance/	Policy engine + JSONL audit	§4.6
src/pipeline.py	End-to-end orchestrator	§3
contracts/MediaRegistry.sol	Solidity registry (per-item + batched)	§4.3
data/generate_synthetic.py	Synthetic dataset generator	§5.1
experiments/exp1_*.py	Signature & watermark latency	§5.2, Table S2
experiments/exp2_*.py	Watermark quality sweep	§5.3, Table S4
experiments/exp3_*.py	Resource consumption	§5.4, Table S6
experiments/exp4_*.py	Gas cost (per-item vs. batched)	§5.5, Table S7
experiments/exp5_*.py	Detection accuracy 2×2 matrix	§5.6, Table S8
tests/test_pipeline.py	23 unit tests (incl. reproducibility)	§5.7
results/*.csv, *.json	Raw experiment output	Section S2
paper/dpf-paper.tex	Main paper source	—
paper/dpf-supplementary.tex	This document	—

repository at <https://github.com/<USER-PLACEHOLDER>/dpf> (archived at <https://doi.org/10.5281/zenodo.<ZENODO-ID-PLACEHOLDER>>).

Data used in this study (released by us). All input data analysed in Experiments 1–5 are synthetic. They are deterministically regenerated by the released code from the random seeds listed below; bit-exact reproducibility is verified by the `pytest` suite shipped with the artefact.

Table S11: Data used in this study and where to find it.

Artefact	Generator / source	Seed(s)
Detector training images	data/generate_synthetic.py (1/ f + perturbation)	0..999
Detector in-dist test images	data/generate_synthetic.py	1000..1399
Detector cross-dist test images	data/generate_synthetic.py (double-perturbation)	2000..2399
Watermark host images	data/generate_synthetic.py (natural-only)	0..19
Latency benchmark digest	64-byte all-zero buffer (deterministic constant)	—
SHA3-256 hash-chain input	32-byte constant “DPF-2026”	—
Human-loop panel accuracy	$\mathcal{N}(0.90, 0.05)$ clipped to $[0.6, 0.99]$	42
Per-trial latency jitter	log-normal multiplicative ($\sigma = 0.04..0.10$)	42
Raw experiment CSVs	experiments/exp{1,...,5}_*.py	42

Third-party datasets *referenced* (not used in experiments). The following three public datasets are discussed in the literature review (§2 of the main paper) and used as comparative anchors in the discussion sections. They are *not* inputs to our experiments and are not redistributed by us. We list the persistent canonical URLs and access conditions for each so a reader can locate the original source.

Ethics, de-identification, and human-subjects status. This study did not involve human participants, human tissue, or animal subjects. All input data are synthetic; no personally identifiable information (PII) is present in any form. The editor’s de-identification requirement therefore does not apply.

Table S12: Third-party benchmark datasets referenced in this work, with persistent canonical URLs.

Dataset	Persistent URL	Access
FaceForensics++ (Rössler et al., 2019)	https://github.com/ondyari/FaceForensics	Google form gated
DFDC (Dolhansky et al., 2020)	https://ai.meta.com/datasets/dfdc/	Meta AI ToU
(Kaggle mirror)	https://www.kaggle.com/c/deepfake-detection-challenge	Kaggle account
WaveFake (Frank & Schönherr, 2021)	https://zenodo.org/records/5642694	CC-BY-NC-SA 4.0
(DOI)	10.5281/zenodo.5642694	—